

SKILL WEEK12

TASK1: To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

```
rishill10@ASUS16X: ~/Skill12
rishill10@ASUS16X:~$ mkdir Skill12
cd Skill12
rishill10@ASUS16X:~/Skill12$ ls -l | grep ".txt" | wc -l
0
ps aux | grep python | wc -l
3
rishill10@ASUS16X:~/Skill12$ sleep 10 &
sleep 15 &
sleep 20 &
[1] 958
[2] 964
[3] 970
rishill10@ASUS16X:~/Skill12$ jobs
ps
[1]      Running                  sleep 10 &
[2]-    Running                  sleep 15 &
[3]+    Running                  sleep 20 &
  PID TTY          TIME CMD
  870 pts/0    00:00:00 bash
  958 pts/0    00:00:00 sleep
  964 pts/0    00:00:00 sleep
  970 pts/0    00:00:00 sleep
  985 pts/0    00:00:00 ps
rishill10@ASUS16X:~/Skill12$ echo "Job 1 started"; sleep 5; echo "Job 1 finished" &
echo "Job 2 started"; sleep 8; echo "Job 2 finished" &
echo "Job 3 started"; sleep 10; echo "Job 3 finished" &
Job 1 started
[1]      Done                    sleep 10
[4] 996
Job 1 finished
Job 2 started
[4]+    Done                    echo "Job 1 finished"
[2]-    Done                    sleep 15
[3]+    Done                    sleep 20
[1] 1005
Job 2 finished
Job 3 started
[1]+    Done                    echo "Job 2 finished"
[1] 1012
Job 3 finished
[1]+    Done                    echo "Job 3 finished"

rishill10@ASUS16X:~/Skill12$ top
top - 03:53:17 up 57 min, 1 user, load average: 0.06, 0.05, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem : 7773.6 total, 7017.1 free, 581.0 used, 328.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7192.6 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+
    1 root        20   0   24144 15324 11532 S   0.0   0.2   0:00.56
     2 root        20   0   3180   2212  2072 S   0.0   0.0   0:00.00
     7 root        20   0   3180   2044  1940 S   0.0   0.0   0:00.00
    47 root       19  -1   50396 16744 15456 S   0.0   0.2   0:00.16
    79 systemd+    20   0   22540 14588 11972 S   0.0   0.2   0:00.06
    86 root       20   0   35484 12360  9152 S   0.0   0.2   0:00.48
   104 root       20   0   2888   1952  1820 S   0.0   0.0   0:00.03
   105 root       20   0   4472   3124  2876 S   0.0   0.0   0:00.00
   106 message+   20   0   8816   5352  4676 S   0.0   0.1   0:00.05
   141 root       20   0   44092 29708 14568 S   0.0   0.4   0:00.19
   162 root       20   0   18436  9392  8140 S   0.0   0.1   0:00.07
   180 syslog     20   0  220548  5584  4656 S   0.0   0.1   0:00.05
   203 root       20   0   5492   2692  2508 S   0.0   0.0   0:00.00
   217 _chrony     20   0   20420  7044  6240 S   0.0   0.1   0:00.04
   223 root       20   0  123456 32592 17984 S   0.0   0.4   0:00.13
   233 _chrony     20   0   12092  2340  1696 S   0.0   0.0   0:00.00
   319 root       20   0   8644   5484  4644 S   0.0   0.1   0:00.01
   363 rishill+   20   0   22336 13540 10916 S   0.0   0.2   0:00.08
   366 rishill+   20   0   22908  4084  2084 S   0.0   0.1   0:00.00
   392 rishill+   20   0   6136   5436  3864 S   0.0   0.1   0:00.02
   865 root       20   0   3184   1124   980 S   0.0   0.0   0:00.00
   866 root       20   0   3200   1260  1108 S   0.0   0.0   0:00.00
   870 rishill+   20   0   6784   6076  3872 S   0.0   0.1   0:00.05
  1028 rishill+   20   0   8184   6028  3916 R   0.0   0.1   0:00.01
```

TASK2: To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```
rishill10@ASUS16X: ~/Skill12  ×  +  v
GNU nano 8.7.1 doc
# Project Architecture

## Components
1. User
2. Application
3. Process Manager
4. Operating System
5. File System

## Execution Flow
User
|
v
Application
|
v
Process
|
v
System Call
|
v
Linux Kernel
|
v
Hardware|

rishill10@ASUS16X: ~/Skill12
GNU nano 8.7.1
+-----+
|  Start  |
+-----+
|         |
v         |
+-----+
| Read Data |
+-----+
|         |
v         |
+-----+
| Process  |
|   Data   |
+-----+
|         |
v         |
+-----+
| Check    |
| Errors   |
+-----+
|         |
v         |
+-----+
| Output   |
+-----+
|         |
v         |
+-----+
|   End    |
+-----+
```

```
rishill10@ASUS16X: ~/Skill12  ×  +  v
GNU nano 8.7.1 docs/execution_path
# Execution Path
1. User starts the application.
2. Linux creates a process.
3. The process executes instructions.
4. The program requests resources from Linux.
5. Linux handles the system call.
6. The program receives the result.
7. The program produces output.
8. The process terminates.|
```

```
rishill10@ASUS16X: ~/Skill12  ×  +  ▾
GNU nano 8.7.1 docs/design/design_decisions.md *
# Design Decisions

## Decision 1
We use Linux processes to execute independent jobs.

## Decision 2
Pipelines are used to transfer output between commands.

## Decision 3
Logs are stored for detecting failures.

## Decision 4
System resources are monitored using Linux utilities.

## Decision 5
Git is used for source code management.
```

```
rishill10@ASUS16X: ~/Skill12  ×  +  ▾  -  □  ×
mkdir docs/design
tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
rishill10@ASUS16X:~/Skill12$ nano docs/architecture.md
rishill10@ASUS16X:~/Skill12$ nano docs/diagrams/flow.txt
rishill10@ASUS16X:~/Skill12$ nano docs/execution_path.md
rishill10@ASUS16X:~/Skill12$ nano docs/design/design_decisions.md
rishill10@ASUS16X:~/Skill12$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
rishill10@ASUS16X:~/Skill12$ find docs -type f
cat docs/architecture.md
docs/diagrams/flow.txt
docs/design/design_decisions.md
docs/architecture.md
docs/execution_path.md
# Project Architecture

## Components
1. User
2. Application
3. Process Manager
4. Operating System
5. File System

## Execution Flow
User
|
v
Application
|
v
Process
|
v
System Call
|
v
Linux Kernel
|
v
Hardware
rishill10@ASUS16X:~/Skill12$ |
```

TASK3: To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

```
rishill10@ASUS16X: ~/Skill12
GNU nano 8.7.1 src/main.c *
#include <stdio.h>

int main()
{
    printf("Project executed successfully!\n");
    return 0;
}
```

```
rishill10@ASUS16X: ~/Skill12
GNU nano 8.7.1
#!/bin/bash

echo "Building project..."

mkdir -p build

gcc src/main.c -o build/main

if [ $? -eq 0 ]; then
    echo "Build successful!"
else
    echo "Build failed!"
    exit 1
fi
```

```
rishill10@ASUS16X: ~/Skill12
GNU nano 8.7.1 README.md *
# Linux Project

## Description
This project demonstrates Linux processes, pipelines, system calls, monitor>

## Directory Structure
- `src` - Source code
- `tests` - Test files
- `scripts` - Utility scripts
- `docs` - Documentation
- `build` - Compiled files

## Build
Run:
`./build.sh`

## Execute
Run:
`./build/main`
```

```
rishill10@ASUS16X: ~/Skill12
rishill10@ASUS16X:~/Skill12$ mkdir -p src tests scripts
tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
rishill10@ASUS16X:~/Skill12$ nano src/main.c
rishill10@ASUS16X:~/Skill12$ nano build.sh
rishill10@ASUS16X:~/Skill12$ chmod +x build.sh
./build.sh
./build/main
Building project...
Build successful!
Project executed successfully!
rishill10@ASUS16X:~/Skill12$ nano README.md
rishill10@ASUS16X:~/Skill12$ git config --global user.email "your_email@example.com"
git config --global user.name "Your Name"
rishill10@ASUS16X:~/Skill12$ git init
git status
git add .
git status
git commit -m "Initial project setup"
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName fal
se"
Initialized empty Git repository in /home/rishill10/Skill12/.git/
On branch master

No commits yet

Untracked files:
```

```
rishill10@ASUS16X: ~/Skill12  x  +  v  -  □  x

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md
  build.sh
  build/
  docs/
  src/

nothing added to commit but untracked files present (use "git add" to track)
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
  new file:   README.md
  new file:   build.sh
  new file:   build/main
  new file:   docs/architecture.md
  new file:   docs/design/design_decisions.md
  new file:   docs/diagrams/flow.txt
  new file:   docs/execution_path.md
  new file:   src/main.c

[master (root-commit) 808c36c] Initial project setup
 8 files changed, 121 insertions(+)
 create mode 100644 README.md
 create mode 100755 build.sh
 create mode 100755 build/main
 create mode 100644 docs/architecture.md
 create mode 100644 docs/design/design_decisions.md
 create mode 100644 docs/diagrams/flow.txt
 create mode 100644 docs/execution_path.md
 create mode 100644 src/main.c
rishill10@ASUS16X:~/Skill12$ git log
git status
commit 808c36c7671ce1effb3a78c9a42432f8a83837a4 (HEAD -> master)
Author: Your Name <your_email@example.com>
Date:   Mon Sep 28 04:00:57 2026 +0000

    Initial project setup
On branch master
nothing to commit, working tree clean
rishill10@ASUS16X:~/Skill12$ |
```

TASK4: To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.

```
rishill10@ASUS16X: ~/Skill12
rishill10@ASUS16X:~/Skill12$ ls -l | grep ".txt" | wc -l
0
rishill10@ASUS16X:~/Skill12$ ps aux | grep python | wc -l
3
rishill10@ASUS16X:~/Skill12$ sleep 100 &
jobs
# Note the PID, e.g., 524
kill -STOP 524
jobs
kill -CONT 524
jobs
kill -9 524
[1] 1388
[1]+  Running                  sleep 100 &
-bash: kill: (524) - No such process
[1]+  Running                  sleep 100 &
-bash: kill: (524) - No such process
[1]+  Running                  sleep 100 &
-bash: kill: (524) - No such process
rishill10@ASUS16X:~/Skill12$ ./build.sh
./build/main
Building project...
Build successful!
Project executed successfully!
rishill10@ASUS16X:~/Skill12$ tree
cat README.md
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
# Linux Project
```

```
## Description
This project demonstrates Linux processes, pipelines, system calls, monitoring and performance analysis.

## Directory Structure
- `src` - Source code
- `tests` - Test files
- `scripts` - Utility scripts
- `docs` - Documentation
- `build` - Compiled files

## Build
Run:
`./build.sh`

## Execute
Run:
`./build/main`
rishill10@ASUS16X:~/Skill12$ git log --oneline
808c36c (HEAD -> master) Initial project setup
rishill10@ASUS16X:~/Skill12$ |
```